Formalizing Fermat, Lecture 10

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Recall group cohomology

Given an abstract group G acting on an abelian group M (or R-module M) we get cohomology groups (or cohomology R-modules) $H^n(G, M)$ for $n \in \mathbb{Z}_{\geq 0}$.

The $H^n(G, M)$ can either be defined as derived functors (or universal delta functors), or as cocycles over coboundaries.

The advantage of cocycles over coboundaries is that it's pretty easy to work out H^1 and maybe even H^2 explicitly, if you have a good understanding of G.

Example: if G acts trivially on M then a 1-cocycle is just $\phi : G \to M$ satisfying $\phi(gh) = \phi(g) + \phi(h)$, so it's just a group homomorphism $G \to M$.

And $gm = m$ so one-coboundaries are trivial, meaning $H^1(G, M) = \mathrm{Hom}(G, M)$ the group homomorphisms.

Note: if you do cocycles and coboundaries then you can't "work out" the formula for the boundary maps – this is a convention.

Restriction

Given that you all seem to know about group cohomology, I'll just note that if $K \subseteq G$ is a subgroup then restricting n-cocycles/n-coboundaries from G^n to K^n works fine and gives maps $H^n(G, M) \to H^n(K, M)$.

(in Lean the M on the right is replaced by $F(M)$ where F is an invisible-to-mathematicians function sending G-modules to K-modules by restricting the action.)

Galois cohomology

For FLT we want to take cohomology of Galois groups. However Galois groups have a topology, and we'd like to keep track of it.

For example we would like to only consider continuous cocycles and coboundaries. But there is a technical subtlety here.

Say M, N, P are topological abelian groups with a continuous action of a topological group G, and $0 \to M \to N \to P \to 0$ is exact.

There's no reason for there to exist a *continuous* section $P \to N$, even though $N \to P$ is surjective.

(example: $0 \to \mathbb{Z} \to \mathbb{R} \to S^1 \to 0$.)

So the boundary homomorphisms (defined by diagram chasing) might not send a continuous n-cocycle for P to a continuous $(n+1)$-cocycle for M.

In particular, the naive approach ("topological groups, topological modules, continuous cocycles and coboundaries") is *not* a derived functor.

Galois cohomology

We fix this technical issue by restricting the kinds of groups and modules we consider (i.e. look at smaller categories, where things do work)

For our topological groups G we will only consider profinite groups, i.e., projective limits of finite groups.

And for our G-modules we will only consider discrete G-modules, i.e. modules M with the discrete topology.

One checks that a G-action on M induces a continuous map $G \times M \to M$ iff the G-stabiliser of each $m \in M$ is open.

Because we don't allow non-trivial topologies on the modules, any surjection $N \to P$ has a continuous splitting (all maps are continuous).

We can then define profinite cohomology, or Galois cohomology, to be continuous cocycles over continuous coboundaries.

Let me explain conceptually why we still get long exact sequences from short exact sequences.

Direct limit approach

Say G is profinite and M is discrete.

I claim that $H^n(G, M) := \varinjlim_{N \leq G} H^n(G/N, M^N)$ where N runs through the open normal subgroups of G. Note that I am using the annoying convention of writing H^n both for group cohomology (on the right) and profinite cohomology (on the left).

First note that a continuous n-cocycle is a locally constant function from G^n to M and takes only finitely many values (image of compact is compact so finite).

Now cover G^n by choosing for each element $(g_i) \in G^n$ an open normal subgroup N_{g_i} of G so that the cocycle is constant on the coset $(g_i)(N_{g_i})^n$.

The $(g_i)(N_{g_i})^n$ cover, so there's a finite subcover, and you take the intersection of the finitely many N_{g_i} involved.

You then intersect this with the intersection of the stabilisers of the image, getting an open subgroup N, and then the cocycle factorizes through a map $(G/N)^n \to M^N$.

Direct limits preserve long exact sequences, so we're done.

Example of Galois cohomology groups

If G is a profinite group acting trivially on M then $H^1(G, M)$ is the continuous group homomorphisms $G \to M$.

Example: say K is a field, and $G = \mathrm{Gal}(\overline{K}/K)$.

Then $H^1(G, \mathbb{Z}/2\mathbb{Z})$ is just the continuous group homomorphisms from G to $\mathbb{Z}/2\mathbb{Z}$.

The nontrivial homomorphisms are determined by their kernel, which is an index 2 normal subgroup of G, so corresponds by Galois theory to a separable quadratic extension of K.

Examples

If K has characteristic not equal to 2, then a quadratic extension of K is just of the form $K(\sqrt{d})$ with $d \in K^\times$.

Note that $K(\sqrt{d}) = K(\sqrt{e})$ if $e = \lambda^2 d$. And Kummer theory states the converse.

So one might expect that $H^1(G, \mathbb{Z}/2\mathbb{Z}) \cong K^\times/(K^\times)^2$.

And indeed that is the case, because squaring is a surjection $\overline{K}^\times \to \overline{K}^\times$, with kernel ± 1 which is isomorphic to $\mathbb{Z}/2\mathbb{Z}$.

So $H^1(G, \mathbb{Z}/2\mathbb{Z}) \cong K^\times/(K^\times)^2$ by a diagram chase (a good exercise)!

You'll need that $H^1(G, \overline{K}^\times) = 0$ ("Hilbert theorem 90", due to Noether).

"Dimension"

Here's something I didn't understand for decades after learning group and Galois cohomology.

We proved last week via an explicit calculation that $H^1(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), i\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$.

So the cyclic group of order 2 has got some cohomology in degree 1.

However there is a periodicity theorem in group cohomology.

It says that if G is a finite cyclic group then
$H^1(G, M) \cong H^3(G, M) \cong H^5(G, M) \cong \cdots$ and
$H^2(G, M) \cong H^4(G, M) \cong H^6(G, M) \cong \cdots$.

In particular, $H^d(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), i\mathbb{Z}) \neq 0$ for all positive odd d.

In topological cohomology theories, we are used to $H^n(X)$ vanishing when n is bigger than the dimension of X.

So this seems to say that the cyclic group of order 2 is somehow infinite-dimensional. For years this surprised me.

Classifying spaces

Here's an explanation.

A group G has a *classifying space* BG.

This is a space with fundamental group G and vanishing higher homotopy groups.

Turns out that the group cohomology $H^n(G, M)$ of G is canonically isomorphic to the singular cohomology of a sheaf $\widetilde{M}$ on the classifying space BG.

Example: $H^n(G, \mathbb{Z})$ is the usual singular cohomology $H^n(BG) = H^n(BG, \mathbb{Z})$.

(If you want to be even fancier, BG is an Eilenberg–Mac Lane space $K(G, 1)$.)

Topologists make the classifying space BG as the quotient of a contractible space EG by a free G-action.

You make the sheaf $\widetilde{M}$ as $(EG \times M)/G$ with G acting diagonally.

Classifying spaces

So what's the classifying space of $\mathbb{Z}/2\mathbb{Z}$?

This sounds like a reasonable question, but the answer is complex.

The group $\mathbb{Z}/2\mathbb{Z}$ acts naturally on the real n-sphere S^n with quotient $\mathbb{R}P^n$, projective space.

Unfortunately S^n isn't EG because it's not contractible.

The "$(n+1)$-dimensional hole" in S^n can however be contracted if you embed S^n into S^{n+1}, and so $EG = \varinjlim_n S^n =: S^\infty$.

Thus $BG = \mathbb{R}P^\infty$, which is indeed infinite-dimensional.

Classifying spaces

Our instinct with groups is that the finite ones are the simplest, but this is not true here.

For group cohomology, the simplest groups are things like $\mathbb{Z}^n$.

Because $E(\mathbb{Z}^n)$ is just $\mathbb{R}^n$ with $\mathbb{Z}^n$ acting in the obvious way, so $B(\mathbb{Z}^n) = (\mathbb{R}/\mathbb{Z})^n$, the n-torus, which is finite-dimensional.

So one would expect $H^d(\mathbb{Z}^2, \mathbb{Z}) = 0$ if $d > 2$ for example, which is true.

More generally, if you know about cohomology of a torus, you'll now know that $H^k(\mathbb{Z}^n, \mathbb{Z}) = \wedge^k(\mathbb{Z}^n)$.

In particular, it vanishes for $k > n$, so our concept of "dimension" is holding out much better.

Galois cohomology

Mazur made the following incredible observations.

If K is a finite extension of $\mathbb{Q}_p$ and $G = \mathrm{Gal}(\overline{K}/K)$ then the Galois cohomology of G behaves like the cohomology of a compact orientable 2-manifold.

And if K is a number field, S is a finite set of finite places of K, if K^S is the maximal extension of K unramified outside $S\infty$, then the profinite cohomology of $\mathrm{Gal}(K^S/K)$ behaves like the cohomology of a 3-manifold with boundary.

This non-existing 3-manifold has one boundary component for each $v \in S$ and one boundary component for each infinite place.

The components for finite places in S are tubular neighbourhoods of knots which have been removed from this manifold.

Technical point: Artin and Mazur associated an *etale homotopy type* to the relevant scheme in these situations ($\mathrm{Spec}(K)$ in the local case, $\mathrm{Spec}(\mathcal{O}_K[1/S])$ in the global case), so not an actual manifold, but the analogy is useful regardless of these technical details.

The local case

Let K be a finite extension of $\mathbb{Q}_p$, and let $G = \mathrm{Gal}(\overline{K}/K)$. Here are some actual facts, rather than analogies.

Let M be a finite G-module (remember that all our G-actions are continuous, so open stabilisers).

Then I claim $H^i(G, M) = 0$ for $i > 2$ (so "dimension of G is 2").

Furthermore $H^i(G, M)$ is finite for all i.

Now let $M^* := \mathrm{Hom}(M, \overline{K}^\times)$ with Galois action $(g \bullet f)(m) := g \bullet (f(g^{-1} \bullet m))$.

Note that M^* is also finite, as it's just $\mathrm{Hom}(M, \mu_N)$ for any positive integer N with $NM = 0$ (e.g., N could be the order of M, or N could be ℓ if M is a $\mathbb{Z}/\ell\mathbb{Z}$-vector space, which it will be in all applications to FLT).

Local Tate duality

G is the absolute Galois group of K, a finite extension of $\mathbb{Q}_p$, M a finite discrete G-module.

I claim that $H^2(G, \overline{K}^\times) = \mathbb{Q}/\mathbb{Z}$ canonically (this is one of the main theorems of local class field theory, and an analogue of $H^2(X, \mathbb{Z}) = \mathbb{Z}$ for an orientable surface).

Local Tate duality is the analogue of Poincaré duality.

It says that for $0 \leq i \leq 2$ there are canonical perfect pairings $H^i(G, M) \times H^{2-i}(G, M^*) \to H^2(G, \overline{K}^\times) = \mathbb{Q}/\mathbb{Z}$ (induced by the cup product).

Alternatively you can imagine the pairing taking values in $H^2(G, \mu_N) = \mathbb{Z}/N\mathbb{Z}$.

The pairings are perfect so in particular the two cohomology groups have the same (finite) size.

In particular, we can work out the size of $H^2(G, M)$ by working out the size of $H^0(G, M^*)$ (which is typically easy).

Local Euler characteristic

Again let M be a finite G-module.

Let's define $\chi(G, M) := |H^0(G, M)||H^2(G, M)|/|H^1(G, M)|$.

If M is a vector space over a finite field k then this is just $\#k$ to the power of $\dim_k(H^0(G, M)) - \dim_k(H^1(G, M)) + \dim_k(H^2(G, M))$, the usual Euler characteristic.

Tate's local Euler–Poincaré characteristic formula says that if $n = |M|$ then $\chi(G, M) = |n|_K$, the norm of $n \in K$, normalised so that $|p|_K = p^{-[K:\mathbb{Q}_p]}$.

If M is a vector space over a finite field k then a clearer way of saying this is $\sum_i (-1)^i \dim_k(H^i(G, M)) = 0$ if $\ell \neq p$, the residue characteristic of K, and if $\ell = p$ it's $-[K : \mathbb{Q}_p] \dim_k M$.

This gives us a formula for the size of $H^1(G, M)$ (assuming we know the sizes of H^0 and H^2, which are typically easy).

Moral: if $K/\mathbb{Q}_p$ is finite and $G = \mathrm{Gal}(\overline{K}/K)$ and M is a finite discrete G-module, we can usually easily work out the order of $H^i(G, M)$ for all i.

Global Galois groups

Now say K is a number field, S is a finite set of finite places of K, and let K^S denote the maximal extension of K unramified outside S and the infinite places.

We want to make rigorous claims backing up the idea that "the cohomology of $G_S := \mathrm{Gal}(K^S/K)$ looks like the cohomology of a 3-manifold with boundary".

But in this situation we would *not* expect to be able to work out sizes of cohomology groups "for free", because (for example) class groups are cohomology groups.

Indeed if S is empty then G_S^{ab} is the Galois group of the maximal abelian extension of K unramified at all finite places, which is the (narrow) class group of K.

So if $N \geq 1$ and $M = \mathbb{Z}/N\mathbb{Z}$ with trivial action then $H^1(G_S, \mathbb{Z}/N\mathbb{Z}) = \mathrm{Hom}(G_S, \mathbb{Z}/N\mathbb{Z})$ is the dual of the N-torsion in the class group, whose order is "mysterious".

Global Galois groups

K a number field, S a finite set of finite places of K, K^S the maximal extension unramified outside $S\infty$.

For simplicity, let's now fix a finite field k of characteristic $\ell > 2$ and stick to M a finite k-vector space.

One reason this makes things a bit simpler is that in the analogy, some of the boundary components of the 3-manifold correspond to the infinite places.

However $H^i(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), M) = 0$ for all $i > 0$ if $\ell > 2$, so this part of the boundary only contributes an H^0.

If $\ell = 2$ then $H^i(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), M)$ can be nonzero for $i = 3, 5, \ldots$

Let's also assume S contains all the primes of K above ℓ.

This will always be true for us, because "unramified at ℓ" is not the right condition in applications, it's "flat at ℓ".

Poitou–Tate

Say M is a finite k-vector space with an action of G_S.

Again we have that $H^i(G_S, M)$ is finite for all i, and zero for $i > 2$ (because $\ell > 2$).

Don't be misled though – vanishing of H^3 doesn't mean that the manifold we're analogising is 2-dimensional.

It's 3-dimensional, but it has a nonempty boundary.

Say $M^* = \mathrm{Hom}_{\mathbb{Z}/\ell\mathbb{Z}}(M, \mu_\ell) = \mathrm{Hom}_k(M, k(\chi))$ is the dual module.

Fact (known in 60s): we have a Poitou–Tate long exact sequence:

$$\begin{aligned} 0 \to H^0(G_S, M) \to \bigoplus_{P \in S} H^0(K_P, M) \to H^2(G_S, M^*)^\vee \to \\ \to H^1(G_S, M) \to \bigoplus_{P \in S} H^1(K_P, M) \to H^1(G_S, M^*)^\vee \to \\ \to H^2(G_S, M) \to \bigoplus_{P \in S} H^2(K_P, M) \to H^0(G_S, M^*)^\vee \to 0. \end{aligned}$$

Poitou–Tate long exact sequence

$$0 \to H^0(G_S, M) \to \bigoplus_{P \in S} H^0(K_P, M) \to H^2(G_S, M^*)^\vee \to$$
$$\to H^1(G_S, M) \to \bigoplus_{P \in S} H^1(K_P, M) \to H^1(G_S, M^*)^\vee \to$$
$$\to H^2(G_S, M) \to \bigoplus_{P \in S} H^2(K_P, M) \to H^0(G_S, M^*)^\vee \to 0.$$

Note that all of these groups are finite.

This is supposed to be some kind of analogue of a Lefschetz duality sequence for compact 3-manifolds with boundary.

The maps between the local and global cohomology groups are just restriction, or the dual of restriction using local Tate duality.

The other two arrows come from some technical diagram chase which I don't understand conceptually (a pairing on subsets $Ш^1$ and $Ш^2$ of H^1 and H^2).

Tate's global Euler characteristic formula

Again K is a number field, k a finite field of characteristic $\ell > 2$ a prime, S a finite set of finite places of K, M a finite k-module with an action of G_S.

The topological Euler characteristic formula for a 3-manifold X with boundary ∂X relates $\chi(X)$ to $\chi(\partial X)$.

The analogue here is that if $\chi(M) := \dim_k(H^0(G_S, M)) - \dim_k(H^1(G_S, M)) + \dim_k(H^2(G_S, M))$ then $\chi(M) = -\sum_{v|\infty} \dim_k(M/M^{G_v})$.

Here G_v is the local Galois group (cyclic of order 2) associated to the map $v : K \to \mathbb{R}$.

This gives us an equation relating $H^1(M)$ and $H^2(M)$.

In the global case, this just means that these two groups are as complicated as each other.

Local conditions

We have been looking at $\mathrm{Gal}(K^S/K)$, where K^S is unramified outside $S\infty$.

The reason for this is that the full cohomology groups $H^i(\mathrm{Gal}(\overline{K}/K), M)$ can be huge even if M is finite.

Example: $H^1(\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}), \mathbb{Z}/2\mathbb{Z}) \cong (\mathbb{Q}^\times)/(\mathbb{Q}^\times)^2$ which is an infinite-dimensional vector space over $\mathbb{Z}/2\mathbb{Z}$ with basis -1 and the primes.

Using $H^i(G_S, M)$ is the same thing as demanding that the cocycles being used to compute the cohomology groups are *unramified* at all finite places outside S.

In other words, their restrictions to $H^i(\mathrm{Gal}(\overline{K}_P/K_P), M)$ are in the image of $H^i(\mathrm{Gal}(K_P^{nr}/K_P), M^{I_P})$.

Here $K_P^{nr} \subset \overline{K}_P$ is the maximal unramified extension of K, and $I_P = \mathrm{Gal}(\overline{K}_P/K_P^{nr})$.

But in practice we have conditions even at the bad primes (for example "flat at ℓ" and "inertia is upper triangular unipotent at P" if $P \in S$ and $P \nmid \ell$).

So we want to put conditions on our cocycles there too.

Local conditions

Say K is a finite extension of $\mathbb{Q}_p$ for a second (it has been a number field for a while now).

Say M is a finite discrete G_K-module and a k-vector space, with k of characteristic ℓ.

Local Tate duality gives an isomorphism $H^1(G_K, M) \cong \mathrm{Hom}_{\mathbb{Z}/\ell\mathbb{Z}}(H^1(G_K, M^*), \mathbb{Z}/\ell\mathbb{Z})$.

Those $\mathbb{Z}/\ell\mathbb{Z}$-linear maps naturally identify with $\mathrm{Hom}_k(H^1(G_K, M^*), k)$ (compose a k-linear map with $\mathrm{tr} : k \to \mathbb{Z}/\ell\mathbb{Z}$).

So we get a k-linear pairing $H^1(G_K, M) \times H^1(G_K, M^*) \to k$.

If $L \subseteq H^1(G_K, M)$ is a k-subspace then we define $L^\perp$ to be its annihilator in $H^1(G_K, M^*)$ under the pairing.

Remark: if $p \neq \ell$ and the action of G_K on M is unramified and L is the unramified cocycles, then $L^\perp$ is the unramified cocycles.

Now let's define the actual cohomology groups we care about.

Selmer groups

As ever, K is a number field, S a finite set of finite places of K, k is a finite field of characteristic $\ell > 2$ and M is a finite-dimensional k-vector space with an action of G_S.

The new data: for each $P \in S$ we choose a subspace $L_P \subseteq H^1(\mathrm{Gal}(\overline{K}_P/K_P), M)$, and write $\mathcal{L} := \{L_P : P \in S\}$.

We let $L_P^\perp$ denote the annihilator of L_P in $H^1(\mathrm{Gal}(\overline{K}_P/K_P), M^*)$.

We define $H^1_{\mathcal{L}}(G_S, M)$ to be the subspace of $H^1(G_S, M)$ consisting of elements whose restriction to $H^1(\mathrm{Gal}(\overline{K}_P/K_P), M)$ lies in L_P for all $P \in S$.

Similarly $H^1_{\mathcal{L}^\perp}(G_S, M^*)$ is the elements which are locally in $L_P^\perp$.

These local changes to the H^1s are "controlled by local terms".

In other words, $H^1_{\mathcal{L}}$ is "no more or less mysterious" as H^1 in the global case.

The key formula

What we need from Galois cohomology, is the following fact.

Say K, S, k, M are as usual.

Then $\dim_k(H^1_{\mathcal{L}}(G_S, M)) - \dim_k(H^1_{\mathcal{L}^\perp}(G_S, M^*)) = \dim_k H^0(G_S, M) - \dim_k H^0(G_S, M^*) - \sum_{v|\infty} \dim_k M^{G_v} + \sum_{P \in S}(\dim_k L_P - \dim_k H^0(\mathrm{Gal}(\overline{K}_P/K_P), M))$.

The proof of this is just an elementary calculation using the profound facts which I've stated.

Although it's easily deducible from 1980s facts, I know of no 1980s reference, so here's how the proof goes.

The key formula

The beginning of the Poitou–Tate long exact sequence is

$$0 \to H^0(G_S, M) \to \bigoplus_{P \in S} H^0(K_P, M) \to H^2(G_S, M^*)^\vee \xrightarrow{\delta}$$
$$\to H^1(G_S, M) \to \bigoplus_{P \in S} H^1(K_P, M) \to H^1(G_S, M^*)^\vee.$$

That δ map from the dual of the global H^2 to the global H^1 has image (by exactness) equal to the global cohomology classes which are *zero* locally at all $P \in S$.

So the image of δ is certainly in $H^1_{\mathcal{L}}(G_S, M)$.

So this is exact too:

$$0 \to H^0(G_S, M) \to \bigoplus_{P \in S} H^0(K_P, M) \to H^2(G_S, M^*)^\vee \xrightarrow{\delta}$$
$$\to H^1_{\mathcal{L}}(G_S, M) \to \bigoplus_{P \in S} L_P \to H^1(G_S, M^*)^\vee.$$

The key formula

$$0 \to H^0(G_S, M) \to \bigoplus_{P \in S} H^0(K_P, M) \to H^2(G_S, M^*)^\vee \xrightarrow{\delta}$$
$$\to H^1_{\mathcal{L}}(G_S, M) \to \bigoplus_{P \in S} L_P \to H^1(G_S, M^*)^\vee.$$

Now by definition of $H^1_{\mathcal{L}^\perp}$ we can complete this exact sequence to

$$0 \to H^0(G_S, M) \to \bigoplus_{P \in S} H^0(K_P, M) \to H^2(G_S, M^*)^\vee \xrightarrow{\delta}$$
$$\to H^1_{\mathcal{L}}(G_S, M) \to \bigoplus_{P \in S} L_P \to H^1(G_S, M^*)^\vee \to H^1_{\mathcal{L}^\perp}(G_S, M^*)^\vee \to 0.$$

This sequence is exact, so the alternating sum of the dimensions of the cohomology groups is zero.

We get $\dim H^1_{\mathcal{L}} - \dim H^1_{\mathcal{L}^\perp}$, a bunch of local terms and H^0s, and then this annoying $\dim H^2(M^*) - \dim H^1(M^*)$.

But we can work this out using Tate's global Euler characteristic, giving the required result.

Conclusion

The moral is: we don't have an easy formula for $\dim_k H^1_{\mathcal{L}}(G_S, M)$, but we do have an easy formula for $\dim_k H^1_{\mathcal{L}}(G_S, M) - \dim_k H^1_{\mathcal{L}^\perp}(G_S, M^*)$.

By "easy" I mean "sums of dimensions of local terms and global H^0s."

Note that the proof uses a bunch of hard results from global class field theory!

My next job is to explain why we care about these cohomology groups.

Deformation theory

I have been constantly saying "oh it's straightforward to get lower bounds for dimensions of deformation rings."

Example: our final boss theorem says that a certain deformation ring is $\mathbb{Z}_\ell$-finite and I've constantly said "this ring is easily checked to have Krull dimension at least 1."

I'm now in a position to explain why this is true.

Recall deformation theory

Recall the set-up: G a profinite topological group, k a finite field, V a finite-dimensional k-vector space, $\overline{\rho} : G \to \mathrm{Aut}_k(V)$ a continuous representation.

A deformation of ρ is a compact Hausdorff local ring R with residue field k, a finite free R-module W with the product topology, and a continuous $\rho : G \to \mathrm{Aut}_R(W)$ such that ρ mod m_R happens to be isomorphic to $\overline{\rho}$.

If $\overline{\rho}$ is absolutely irreducible then this deformation problem is representable, so there's some universal $\rho^{\mathrm{univ}} : G \to \mathrm{GL}_n(R^{\mathrm{univ}})$ such that any deformation $\rho : G \to \mathrm{Aut}_R(W)$ is isomorphic to $\rho^{\mathrm{univ}} \otimes_{R^{\mathrm{univ}}} R$ for some $R^{\mathrm{univ}} \to R$.

Variants

There are a couple of variants we'll need.

Firstly, we might want to consider only deformations which have some properties (e.g. flat, cyclotomic determinant), and if the property satisfies some simple axioms then the deformation problem is again representable.

Secondly, we might fix $\mathcal{O}$ the integers in a finite extension of $\mathbb{Q}_\ell$ with residue field k and only consider deformations to $\mathcal{O}$-algebras.

This is a "soft" change – this is still representable, you just tensor up the universal deformation ring from $W(k)$ (the Witt vectors of k) to $\mathcal{O}$.

Let me also explain another "soft" change – where we consider "lifts" instead of deformations.

Lifts

Say $\overline{\rho} : G \to \mathrm{GL}_n(k)$.

A *lift* of $\overline{\rho}$ is $\rho : G \to \mathrm{GL}_n(R)$ for some compact Hausdorff local R with residue field k, such that ρ mod m_R equals $\overline{\rho}$.

How is this different from deformations?

Well distinct lifts can be abstractly *isomorphic* without being *equal*.

For example if M is in $\mathrm{GL}_n(R)$ and if M is the identity when reduced mod m_R, then $M\rho M^{-1}$ is typically a different lift, but the same deformation.

Lifts are more "rigid" than deformations.

Example: universal deformation rings exist when $\overline{\rho}$ is absolutely irreducible.

But the same proof shows that universal lifting rings exist without any irreducibility hypotheses on $\overline{\rho}$.

Lifts vs deformations

Lifts: we actually have the matrix entries.

Deformations: we only consider things up to isomorphism.

To get a lift from a deformation $\rho : G \to \mathrm{Aut}_R(W)$, we need to pick an R-basis of W.

So it feels like we have n^2 extra degrees of freedom.

Another way of thinking about it: we can conjugate a lift by an $n \times n$ matrix M and get a different lift with the same deformation.

But actually, if we scale all basis elements by the same constant factor, we get the same matrices.

Equivalently: if M is a scalar matrix then conjugation doesn't change the lift.

So maybe a better guess is that we have $n^2 - 1$ extra degrees of freedom.

Indeed, that is the case.

Deformations and lifts

Lemma

If $\overline{\rho} : G \to \mathrm{GL}_n(k)$ is absolutely irreducible, with universal deformation $\rho^{\mathrm{univ}} : G \to \mathrm{GL}_n(R^{\mathrm{univ}})$ (choose a basis!), then the universal lifting ring $R^{\square}$ exists and it's isomorphic to $R^{\mathrm{univ}}[[X_{1,1}, X_{1,2}, \ldots, X_{n,n}]]/(X_{1,1})$.

Proof:

Given the universal deformation ρ^{univ}, define $R^{\square} := R^{\mathrm{univ}}[[X_{1,1}, X_{1,2}, \ldots, X_{n,n}]]/(X_{1,1})$ and define the lift $\rho^{\square} : G \to \mathrm{GL}_n(R^{\square})$ by $\rho^{\square}(g) = (\mathbf{1}_n + (X_{i,j}))\rho^{\mathrm{univ}}(g)(\mathbf{1}_n + (X_{i,j}))^{-1}$.

We just need to check that, given a lift ρ to $\mathrm{GL}_n(R)$, there's a unique map from $R^{\square}$ to R which realises this lift.

The lift gives a deformation, so we get a unique map $R^{\mathrm{univ}} \to R$.

We then know that $\rho^{\mathrm{univ}} \otimes_{R^{\mathrm{univ}}} R \cong \rho$.

So there's some matrix M such that $M(\rho^{\mathrm{univ}} \otimes_{R^{\mathrm{univ}}} R)M^{-1} = \rho$.

Deformations and lifts

We have M such that $M(\rho^{\mathrm{univ}} \otimes_{R^{\mathrm{univ}}} R)M^{-1} = \rho$.

One checks that absolute irreducibility of $\overline{\rho}$ implies that M is well-defined up to scalars.

Reducing mod m_R we deduce $(M \bmod m_R)\overline{\rho}(M \bmod m_R)^{-1} = \overline{\rho}$.

Because $\overline{\rho}$ is absolutely irreducible, $(M \bmod m_R)$ is a scalar in $\mathrm{GL}_n(k)$.

In particular, its top left entry reduces to a unit, so is a unit in R.

So we can nail M down uniquely by scaling it so that the top left entry is 1.

Now $M = \mathbf{1}_n + (m_{i,j})$ with $m_{1,1} = 0$ and $m_{i,j} \in m_R$.

So there's a unique map $R^{\square} \to R$ extending our map $R^{\mathrm{univ}} \to R$ and sending X_{ij} to $m_{i,j}$, making $\rho^{\square} \otimes_{R^{\square}} R = \rho$.

This is what we needed to show.

Infinitesimal deformations

As ever, G is a profinite group and $\overline{\rho} : G \to \mathrm{GL}_n(k)$ is a continuous representation.

Say $\rho : G \to \mathrm{GL}_n(R)$ is a lift of $\overline{\rho}$, with m_R the maximal ideal of R.

I claim that there are natural bijections between $(m_R/((m_R)^2, \ell))^\vee$ and $\mathrm{Hom}(R, k[\epsilon]/(\epsilon^2))$ (in our category $\mathcal{C}_k$).

Given $\phi : m_R/((m_R)^2, \ell) \to k$, define $f : R \to k[\epsilon]/(\epsilon^2)$ by first mapping R to R/ℓ, which can be checked to be a k-algebra.

And then writing an element of R/ℓ as $a + m$ with $a \in k$ and $m \in m_R$, and sending it to $a + \phi(m)\epsilon$.

Going the other way is easier: given $f : R \to k[\epsilon]/(\epsilon^2)$ in our category, we have $f(m_R) \subseteq \epsilon k$ (morphisms have to be the identity on residue fields).

So we get a map $\phi : m_R \to k$ such that $\epsilon\phi(x) = f(x)$.

One checks easily that m_R^2 and ℓ are in the kernel.

Lifts and cocycles

When $R = R^\square$ we can go further.

Given $\overline{\rho}$, set $\mathrm{ad}(\overline{\rho}) := \mathrm{Hom}_k(\overline{\rho}, \overline{\rho})$ with $(g \bullet f)(v) = g \bullet (f(g^{-1} \bullet v))$.

I claim that the following four sets naturally biject with one another:

- $(m^\square/((m^\square)^2, \ell))^\vee$;
- $\mathrm{Hom}(R^\square, k[\epsilon]/(\epsilon^2))$ (in our category);
- lifts of $\overline{\rho}$ to $k[\epsilon]/(\epsilon^2)$;
- 1-cocycles $G \to \mathrm{ad}(\overline{\rho})$.

We just saw that the first two naturally biject (for any lift R, not just the universal lift).

The second and third biject by definition of a universal lifting ring.

So we just need to relate these things to the 1-cocycles.

Lifts and cocycles

Say $f : G \to \mathrm{ad}(\overline{\rho})$ is a continuous function.

Define a map $\tilde{\rho} : G \to \mathrm{GL}_n(k[\epsilon])$ by $\tilde{\rho}(g) = \overline{\rho}(g) + \epsilon f(g)(\overline{\rho}(g))$.

Now do the pleasant exercise of checking that $\tilde{\rho}$ is a representation lifting $\overline{\rho}$ iff f is a 1-cocycle.

Conversely, any lift of $\overline{\rho}$ to $k[\epsilon]$ is of the form $\tilde{\rho}$ above, so given $\tilde{\rho}$ we can get an f.

Deformations and cohomology

We have just seen that the dual of $m^\square/(m^\square)^2$ for the lifting ring $R^\square/\ell$ naturally bijects with 1-cocycles.

Now assume $\overline{\rho}$ is absolutely irreducible. So deformations are lifts modulo n^2-1 relations.

1-cohomology classes are 1-cocycles modulo 1-coboundaries, which come from elements of $\mathrm{ad}(\overline{\rho})$, which is linear maps $k^n \to k^n$, so has dimension n^2.

But Galois acts trivially on the scalar maps, so they produce the trivial coboundary. And abs irreducibility of $\overline{\rho}$ means that these are the only Galois-stable elements of $\mathrm{ad}(\overline{\rho})$.

This adds credibility to the idea that if m_{univ} is the maximal ideal of R^{univ}, the universal deformation ring of the absolutely irreducible $\overline{\rho}$, then the dual of $m_{\mathrm{univ}}/(m_{\mathrm{univ}}^2, \ell)$ might naturally biject with $H^1(G, \mathrm{ad}(\overline{\rho}))$.

Again this is a pleasant exercise: conjugating a lift by $1+\epsilon M$ with $M \in M_{n\times n}(k)$ corresponds to changing the cocycle by the coboundary coming from M considered as a map $k^n \to k^n$.